

Quartic Zero-Repair: Positive Packet Transport, Waring Phase Carriers, and Resolvent Boundary Compression

An incomplete preprint toward the logarithmic reserve bound

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Abstract

We study the exact history-indexed zero-repair problem for the explicit quartic pair

$$P = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = \frac{1}{64}(7, 20, 10, 20, 7),$$

whose probability generating functions satisfy

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

The cost $\mathcal{B}_n$ is the minimum expected initial nonnegative reserve required to transform fresh iid Q -inputs into iid P -outputs for n causal steps. We record a self-contained logarithmic lower bound and the unconditional finite-horizon estimate

$$\mathcal{B}_n \leq \frac{473}{56} n^{1/4},$$

as well as a weak- L^1 reduction of the still-open matching logarithmic upper bound, the exact one-fan stochastic-order characterization, and an exact tail-interlacing theorem that preserves the leading $1/j$ tail in all five children.

The main constructive contribution of this preprint is a positive three-regime Q -fresh packet fan. For every parent law with boundary mass $p = B(0) \leq 1/4$, a packet of mass $3p$ can compensate the boundary output defect while leaving public branches 1 and 3 exactly unchanged. A top-tail choice gives explicit low-prefix recursions and the sharp one-step boundary criterion

$$14B(0) + 4B(1) \leq 5.$$

We then pass to a Waring/Beta phase basis in which downward motion at the hard boundary has the exact aging law

$$DW_{0,k} = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}W_{0,k+1}.$$

This produces a positive phase carrier closed under the shift operators appearing in the packet fan. We prove, in addition, an all-depth theorem for the pure public branch 4^m : a two-scalar region in the $W_{0,k}$ -cone is invariant under the greedy low-leak packet rule, so the previously delicate branch-4 boundary accumulation is rigorously controlled for every depth. For mixed histories we first introduce a discounted Waring leak score to keep track of both distance from the hard boundary and phase aging. The decisive refinement is a law-level resolvent functional

$$\mathcal{R}_\rho(M) = \sum_{j \geq 1} \rho^{j-1} M(j), \quad 0 < \rho < 1,$$

for which upward and downward reserve shifts satisfy exact identities rather than one-sided phase majorants. Under the packet fan, the boundary mass, the reserve-one mass, and the

resolvent evolve through an explicit three-dimensional state (p, q, R) , while a packet enters only through two linear controls $(h, g) = (G(1), \mathcal{R}_\rho(G))$. The only remaining second-layer quantities admit universal interval bounds, and every nonnegative linear packet cost has a law-independent proportional-selection bound. Thus the all-history closure problem becomes a finite-dimensional robust invariant-control problem rather than a transport problem on an infinite phase inventory.

If such an invariant control region is constructed from the critical root $H_2 = W_{0,1}$, then

$$\mathcal{B}_n \leq 2H_{4n} = O(\log n),$$

and hence the logarithmic order is sharp. This is intentionally an incomplete preprint: that invariant region, and therefore the all-history packet invariance statement, is not claimed as proved.

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1 The quartic zero-repair problem

Let

$$P = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = \frac{1}{64}(7, 20, 10, 20, 7). \quad (1)$$

Their probability generating functions obey

$$\boxed{G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}}, \quad (2)$$

and both laws have mean 2.

Definition 1.1 (Exact history-indexed tree). For every output history $h \in \{0, 1, 2, 3, 4\}^{\leq n}$, let B_h be a probability law on $\mathbb{N}_0$. The family is an exact depth- n zero-repair tree if at every internal node

$$\boxed{Q * B_h = \sum_{x=0}^4 P_x \delta_x * B_{hx}}. \quad (3)$$

For finite horizon we require the root to have finite first moment.

Definition 1.2 (Reserve bill). Let $\mathcal{K}_n$ denote the set of root laws of exact depth- n trees. Define

$$\boxed{\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}_B J, \quad \mathcal{B}_0 := 0.} \quad (4)$$

The convolution identity is not merely a static relation. It admits a lossless causal implementation with fresh Q -input at every step.

Lemma 1.3 (Lossless causal lift). *Suppose (3) holds at node h , and write $S_h = Q * B_h$. For $S_h(s) > 0$, define*

$$K_h(x \mid s) := \frac{P_x B_{hx}(s-x)}{S_h(s)}, \quad B_{hx}(k) := 0 \quad (k < 0).$$

If $J_t \sim B_h$, draw $Y_{t+1} \sim Q$ independently, then draw X_{t+1} according to $K_h(\cdot \mid J_t + Y_{t+1})$, and put

$$J_{t+1} = J_t + Y_{t+1} - X_{t+1}.$$

Then $J_{t+1} \geq 0$, the public output has law P , the fresh input remains independent of the past, and conditional on $X_{t+1} = x$ the new reserve has law B_{hx} .

Proof. For fixed $x, k \geq 0$,

$$\begin{aligned} \mathbb{P}(X_{t+1} = x, J_{t+1} = k) &= S_h(k+x) K_h(x \mid k+x) \\ &= P_x B_{hx}(k). \end{aligned}$$

Summing over k gives the P -marginal, while division by P_x gives the prescribed child law. The convention $B_{hx}(k) = 0$ for $k < 0$ forces nonnegative reserve. $\square$

Taking first moments in (3) and using $\mathbb{E}P = \mathbb{E}Q$ gives the exact fan conservation law

$$\boxed{\mathbb{E}B_h = \sum_{x=0}^4 P_x \mathbb{E}B_{hx}}. \quad (5)$$

Lemma 1.4 (Finite-support reduction). *For every finite horizon n , the infimum in (4) is attained, and an optimal root may be chosen with support in $\{0, \dots, 4n\}$.*

Proof. Use the causal realization above. For any feasible depth- n trajectory define

$$D = (J_0 - 4n)^+, \quad J'_t = J_t - D.$$

On $\{J_0 > 4n\}$,

$$J'_t \geq 4n - 4t \geq 0,$$

since one step can decrease the reserve by at most 4. Otherwise $D = 0$. The (X, Y) -trajectory is unchanged, so freshness and exactness are preserved, while the root becomes $\min(J_0, 4n)$. The resulting finite occupancy problem is a compact linear program. $\square$

2 A logarithmic lower bound

The quartic defect is small near $z = 1$, but it cannot be repaired with uniformly bounded expected reserve. The following argument is purely finite-horizon.

Theorem 2.1 (Logarithmic reserve growth). *Fix $0 < z < 1$ and write*

$$p := G_P(z), \quad q := G_Q(z), \quad c(z) := \frac{\log(p/q)}{-\log z} > 0.$$

For integers $r \geq t \geq 1$,

$$\boxed{\mathcal{B}_r \geq (1 - p^t)\mathcal{B}_{r-t} + t c(z) p^t.} \quad (6)$$

Consequently

$$\boxed{\mathcal{B}_n \geq \max_{1 \leq t \leq n} t c(z) \left[1 - (1 - p^t)^{\lfloor n/t \rfloor} \right],} \quad (7)$$

and

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\log(G_P(z)/G_Q(z))}{(-\log z)(-\log G_P(z))} > 0.} \quad (8)$$

Proof. Take an optimal depth- r tree. Put

$$F_h(z) := \mathbb{E} z^{J_h}, \quad m_h := \mathbb{E} B_h, \quad b := \mathcal{B}_{r-t}, \quad a := -\log z.$$

Iterating (3) through t levels gives

$$q^t F_\emptyset(z) = p^t \mathbb{E}_{P_z^{\otimes t}} F_g(z), \quad P_z(x) := \frac{P_x z^x}{p}.$$

Jensen's inequality yields $F_g(z) \geq e^{-am_g}$, hence

$$\mathbb{E}_{P_z^{\otimes t}} m_g \geq \frac{t \log(p/q) - \log F_\emptyset(z)}{a} \geq t c(z).$$

Every depth- t descendant is the root of a feasible depth- $(r-t)$ subtree, so $m_g \geq b$. Moreover

$$\frac{P_z^{\otimes t}(g)}{P(g)} = \frac{z^{\sum_i g_i}}{p^t} \leq p^{-t}.$$

Using (5),

$$\begin{aligned} \mathbb{E}_{P_z^{\otimes t}} m_g &= b + \mathbb{E}_{P_z^{\otimes t}} (m_g - b) \\ &\leq b + p^{-t} \sum_{|g|=t} P(g) (m_g - b) \\ &= b + p^{-t} (\mathcal{B}_r - b). \end{aligned}$$

Combining the last two displays gives (6). Iteration over blocks gives (7). Finally choose

$$t_n = \left\lfloor \frac{\log n - 2 \log \log n}{-\log p} \right\rfloor.$$

Then $t_n / \log n \rightarrow 1 / (-\log p)$ and $\lfloor n/t_n \rfloor p^{t_n} \rightarrow \infty$, so the bracket in (7) tends to one. This gives (8). $\square$

At $z = 1/16$,

$$G_P(z) = \frac{74273}{524288}, \quad G_Q(z) = \frac{543559}{4194304},$$

so

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \kappa_0 := \frac{\log(594184/543559)}{\log 16 \log(524288/74273)} = 0.0164347253 \dots \quad (9)$$

3 An unconditional finite-horizon upper bound

The logarithmic upper bound developed later is still conditional. Independently of it, the quartic defect already yields an explicit causal construction with sublinear cost. We include the argument because it is part of the unconditional state of the problem and provides the present all-horizon upper benchmark.

Write

$$\Delta := \delta_0 - \delta_1.$$

Then (2) is equivalent to

$$\boxed{P - Q = \frac{1}{64}\Delta^{*4}.} \quad (10)$$

For $a > 0$, define the geometric law

$$\gamma_a(j) = \frac{1}{1+a} \left(\frac{a}{1+a} \right)^j, \quad \mathbb{E}\gamma_a = a, \quad (11)$$

and let

$$C_a := \gamma_a^{*4}.$$

Put

$$\varepsilon := \frac{25}{7a^3}, \quad E_\varepsilon := (1 - \varepsilon)\delta_0 + \varepsilon\delta_1, \quad (12)$$

assuming $a^3 \geq 25/7$, and for $r \geq 0$ set

$$\boxed{R_r := C_a * E_\varepsilon^{*r}.} \quad (13)$$

Thus

$$\boxed{\mathbb{E}R_r = 4a + r\varepsilon.} \quad (14)$$

Lemma 3.1 (Third-difference insurance inequality). *For every $a > 0$, coefficientwise,*

$$\boxed{\Delta^{*3} * C_a + \frac{25}{a^3}C_a \geq 0.} \quad (15)$$

Consequently

$$\boxed{\Delta^{*3} * R_r \geq -\frac{25}{a^3}R_r} \quad (16)$$

for all $r \geq 0$.

Proof. The geometric series gives

$$\Delta * \gamma_a = \frac{1}{a}(\delta_0 - \gamma_a).$$

Hence

$$\Delta^{*3} * C_a = \frac{1}{a^3}(\gamma_a - 3\gamma_a^{*2} + 3\gamma_a^{*3} - \gamma_a^{*4}).$$

After adding $25C_a/a^3$, divide the coefficient at j by the positive number $\gamma_a(j)$, and write $\delta = (1+a)^{-1}$, $u = \delta(j+1)$. The resulting expression is at least

$$1 - 3u + \frac{3}{2}u^2 + 4u^3 = (u+1)(2u-1)^2 + \frac{3}{2}u^2 \geq 0.$$

This proves (15). Convolution with the positive measure E_ε^{*r} gives (16). $\square$

Lemma 3.2 (Scale-fixed stochastic dominance). *For every $r \geq 1$,*

$$\boxed{Q * R_r \succeq_{\text{st}} P * R_{r-1}.} \quad (17)$$

Proof. Since $R_r = R_{r-1} * (\delta_0 - \varepsilon \Delta)$, (10) gives

$$Q * R_r - P * R_{r-1} = -\frac{1}{64} \Delta^{*4} * R_{r-1} - \varepsilon \Delta * Q * R_{r-1}.$$

For every measure M , $\sum_{j=0}^k (\Delta * M)(j) = M(k)$. Therefore

$$F_{Q * R_r}(k) - F_{P * R_{r-1}}(k) = -\frac{1}{64} (\Delta^{*3} * R_{r-1})(k) - \varepsilon (Q * R_{r-1})(k).$$

Because $Q_0 = 7/64$,

$$(Q * R_{r-1})(k) \geq \frac{7}{64} R_{r-1}(k).$$

Using (16) and $\varepsilon = 25/(7a^3)$, the right-hand side is at most zero. Hence the required stochastic dominance holds. $\square$

Theorem 3.3 (Explicit causal $O(n^{1/4})$ upper bound). *For every $n \geq 1$ and every $a > 0$ with $a^3 \geq 25/7$,*

$$\boxed{\mathcal{B}_n \leq 4a + \frac{25n}{7a^3}.} \quad (18)$$

In particular, with $a = 2n^{1/4}$,

$$\boxed{\mathcal{B}_n \leq \frac{473}{56} n^{1/4}.} \quad (19)$$

Proof. Start from $B_\emptyset = R_n$. Suppose a node with r steps remaining satisfies $B_h \succeq_{\text{st}} R_r$, and set

$$\mu := Q * B_h, \quad \alpha := P * R_{r-1}.$$

By monotonicity of convolution and (17), $\mu \succeq_{\text{st}} \alpha$. Choose a monotone coupling (U, S) of α and μ with $U \leq S$. Conditional on $U = u$, sample X from the canonical decomposition

$$\alpha(u) = \sum_x P_x R_{r-1}(u - x).$$

Then $Z := U - X$, conditional on any $X = x$, has law R_{r-1} , while the actual reserve satisfies

$$J' = S - X \geq U - X = Z.$$

Thus every child law stochastically dominates R_{r-1} , and the public law is exactly P . Iteration yields an exact depth- n tree. By (14), its root cost is

$$4a + n\varepsilon = 4a + \frac{25n}{7a^3}.$$

For $a = 2n^{1/4}$, this equals

$$8n^{1/4} + \frac{25}{56} n^{1/4} = \frac{473}{56} n^{1/4}.$$

The finite-support reduction from Section 1 may then be applied without increasing the mean. $\square$

Combining this theorem with the logarithmic lower bound gives the unconditional sandwich

$$\boxed{\kappa_0 \log n - o(\log n) \leq \mathcal{B}_n \leq \frac{473}{56} n^{1/4}.} \quad (20)$$

The remainder of the paper concerns the still-open improvement of the right-hand side to $O(\log n)$.

4 Weak- L^1 reduction of the logarithmic upper bound

The matching upper bound can be reduced to an infinite exact tree whose root has critical $1/j$ tail.

Theorem 4.1 (Truncated-reserve criterion). *Allow an infinite exact tree whose node laws may have infinite first moment. If $J_0 \sim B_\emptyset$, then for every $n \geq 1$,*

$$\boxed{\mathcal{B}_n \leq \mathbb{E} \min(J_0, 4n) = \sum_{j=1}^{4n} \mathbb{P}(J_0 \geq j).} \quad (21)$$

If

$$\mathbb{P}(J_0 \geq j) \leq \frac{A}{j} \quad (j \geq 1),$$

then

$$\boxed{\mathcal{B}_n \leq AH_{4n} \leq A(1 + \log(4n)).} \quad (22)$$

Proof. For a fixed horizon n , subtract the same random quantity

$$D = (J_0 - 4n)^+$$

from every reserve during the first n steps. If $J_0 > 4n$, then

$$J'_t = 4n + \sum_{i=1}^t (Y_i - X_i) \geq 4n - 4t \geq 0.$$

The public and fresh-input trajectories are unchanged, so the shifted laws form an exact depth- n tree with root $\min(J_0, 4n)$. The rest is the tail-sum identity and the harmonic bound. $\square$

A convenient critical root for the packet construction below is

$$H_2(0) = 0, \quad H_2(j) = \frac{2}{(j+1)(j+2)} \quad (j \geq 1), \quad (23)$$

for which

$$\boxed{\mathbb{P}_{H_2}(J \geq j) = \frac{2}{j+1} \leq \frac{2}{j}.} \quad (24)$$

Thus an infinite exact tree rooted at H_2 would imply

$$\boxed{\mathcal{B}_n \leq 2H_{4n} = 2\log n + O(1).} \quad (25)$$

5 One-step transport and exact tail interlacing

Let B be a parent reserve law and put $S = Y + J$, with $Y \sim Q$, $J \sim B$ independent.

Theorem 5.1 (One-fan equivalence). *A family of probability laws B_x satisfies*

$$Q * B = \sum_{x=0}^4 P_x \delta_x * B_x$$

*if and only if there exists a coupling π of $X \sim P$ and $S \sim Q * B$, supported on $X \leq S$, such that*

$$B_x(j) = \frac{\pi(x, x+j)}{P_x}.$$

Consequently a positive one-fan exists if and only if

$$\boxed{P \preceq_{\text{st}} Q * B.} \quad (26)$$

For (1), writing $b_j = B(j)$, this is equivalent to

$$27b_0 + 7b_1 \leq 24, \quad (27)$$

$$37b_0 + 27b_1 + 7b_2 \leq 40, \quad (28)$$

$$57b_0 + 37b_1 + 27b_2 + 7b_3 \leq 56. \quad (29)$$

Proof. From a fan define $\pi(x, s) = P_x B_x(s - x)$. Conversely, recover B_x from the displayed formula. The order support is exactly nonnegativity of the residual reserve. In one dimension, existence of an order coupling is equivalent to stochastic domination. Since P is supported on $\{0, 1, 2, 3, 4\}$, only the cumulative inequalities at $0, 1, 2, 3$ matter; direct substitution gives (27)–(29). See, e.g., [1, 2]. $\square$

The next theorem separates the infinite tail from the finite boundary transport.

Theorem 5.2 (Strict-core exact-tail interlacing). *Let $\mu = Q * B$ and suppose*

$$F_\mu(k) < F_P(k), \quad k = 0, 1, 2, 3.$$

Then there exists a finite cutoff $H \geq 4$ and an order coupling π such that

$$\boxed{\pi(x, s) = P_x \mu_s \quad (s > H).} \quad (30)$$

Hence every child obeys

$$\boxed{B_x(j) = \mu_{x+j} \quad (x + j > H).} \quad (31)$$

If

$$\mathbb{P}_B(J \geq j) = \frac{A}{j} + O(j^{-2}),$$

then for every x ,

$$\mathbb{P}_{B_x}(J \geq j) = \frac{A}{j} + O(j^{-2}). \quad (32)$$

Proof. Let $T_H = \mu((H, \infty))$ and $c_H = 1 - T_H$. Normalize the truncation by

$$\mu^{(H)}(s) = \frac{\mu_s}{c_H}, \quad 0 \leq s \leq H.$$

For $k = 0, 1, 2, 3$,

$$F_{\mu^{(H)}}(k) = \frac{F_\mu(k)}{c_H} \longrightarrow F_\mu(k) < F_P(k).$$

For all sufficiently large H , $P \preceq_{\text{st}} \mu^{(H)}$. Let ρ_H be an order coupling of these finite laws and set

$$\pi(x, s) = \begin{cases} c_H \rho_H(x, s), & s \leq H, \\ P_x \mu_s, & s > H. \end{cases}$$

This has the required marginals and support. Equation (31) follows immediately. Since Y has bounded support, convolution with Q preserves a tail $A/j + O(j^{-2})$, and a bounded shift by x preserves the same leading coefficient. $\square$

6 A positive three-regime packet fan

We now give an explicit causal coupling that isolates the hard boundary correction into a selectable positive packet.

For a measure M on $\mathbb{N}_0$, define

$$(DM)(j) := M(j+1), \quad (UM)(j) := \begin{cases} 0, & j = 0, \\ M(j-1), & j \geq 1. \end{cases}$$

Consider the three Q -column couplings

$$C^+ = \frac{1}{64} \begin{pmatrix} 7 & 4 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 & 0 \\ 0 & 0 & 10 & 4 & 2 \\ 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}, \quad (33)$$

$$C^0 = \frac{1}{64} \begin{pmatrix} 7 & 1 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 & 0 \\ 0 & 3 & 10 & 3 & 0 \\ 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 1 & 7 \end{pmatrix}, \quad (34)$$

and

$$C^- = \frac{1}{64} \begin{pmatrix} 7 & 0 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 & 0 \\ 0 & 4 & 10 & 2 & 0 \\ 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 2 & 7 \end{pmatrix}. \quad (35)$$

Their row laws are

$$R^+ = P + \frac{3}{64}(\delta_0 - \delta_4), \quad R^0 = P, \quad R^- = P - \frac{1}{64}(\delta_0 - \delta_4). \quad (36)$$

Moreover C^+ is supported on $X \leq Y$, while C^0, C^- are supported on $X \leq Y + 1$.

Theorem 6.1 (Packet fan). *Let B be a probability law on $\mathbb{N}_0$, let $p = B(0) \leq 1/4$, and choose any submeasure*

$$G \leq B - p\delta_0, \quad |G| = 3p. \quad (37)$$

Put

$$A := B - p\delta_0 - G.$$

Use C^+ on the boundary mass $p\delta_0$, C^- on G , and C^0 on A . Then the resulting one-step protocol is causal, has fresh input law Q , public output law exactly P , and nonnegative residual reserve.

Its five child laws are

$$B_0 = \frac{1}{8} [p(7\delta_0 + 4\delta_1) + 7A + UA + 7G], \quad (38)$$

$$B_1 = B, \quad (39)$$

$$B_2 = \frac{1}{16} [p(10\delta_0 + 4\delta_1 + 2\delta_2) + 3DA + 10A + 3UA + 4DG + 10G + 2UG], \quad (40)$$

$$B_3 = B, \quad (41)$$

$$B_4 = \frac{1}{8} [5p\delta_0 + DA + 7A + 2DG + 7G]. \quad (42)$$

Proof. The public row law is

$$pR^+ + |G|R^- + |A|P.$$

Using $|G| = 3p$, $|A| = 1 - 4p$, and (36), the correction in $\delta_0 - \delta_4$ cancels exactly, leaving P . At $J = 0$, C^+ never spends more than the fresh input. On $J \geq 1$, both C^0 and C^- satisfy $X \leq Y + 1$, so one unit of reserve is sufficient. Reading each row and converting $j \mapsto j + Y - X$ gives (38)–(42). In particular the rows $x = 1, 3$ are exact identities in all three couplings. $\square$

The identities

$$\boxed{B_1 = B_3 = B} \tag{43}$$

are central: the two odd public branches transport the complete hidden state without distortion.

7 Top-tail packet selection and exact prefix dynamics

The packet location is a genuine phase degree of freedom. A particularly sharp deterministic choice is to take G from the highest reserve states downward, splitting the threshold atom if needed.

Write

$$p = B(0), \quad q = B(1), \quad r = B(2).$$

For the top-tail packet of mass $3p$, define

$$g_1 := G(1) = (4p + q - 1)_+, \tag{44}$$

and

$$g_2 := G(2) = (4p + q + r - 1)_+ - (4p + q - 1)_+. \tag{45}$$

Proposition 7.1 (Exact low-prefix recursion). *For the top-tail packet fan,*

$$p_0 := B_0(0) = \frac{7}{8}p, \quad q_0 := B_0(1) = \frac{1}{2}p + \frac{7}{8}q, \tag{46}$$

$$p_1 = p_3 = p, \quad q_1 = q_3 = q, \tag{47}$$

$$p_2 = \frac{5}{8}p + \frac{3}{16}q + \frac{1}{16}g_1, \quad q_2 = \frac{1}{4}p + \frac{10}{16}q + \frac{3}{16}r + \frac{1}{16}g_2, \tag{48}$$

$$p_4 = \frac{5}{8}p + \frac{1}{8}q + \frac{1}{8}g_1, \quad q_4 = \frac{7}{8}q + \frac{1}{8}r + \frac{1}{8}g_2. \tag{49}$$

Proof. Equations (44)–(45) are the defining cumulative identities of the highest-mass submeasure of total mass $3p$: the packet reaches state 1 only after exhausting all mass above 1, and similarly for state 2. The formulas then follow by reading the coefficients of δ_0 and δ_1 in (38)–(42). $\square$

The first boundary face collapses to a single linear inequality.

Theorem 7.2 (Sharp one-step boundary criterion for the top-tail packet). *Assume $p \leq 1/4$. Under the top-tail packet fan,*

$$\boxed{B_x(0) \leq \frac{1}{4} \quad \text{for every } x = 0, \dots, 4} \tag{50}$$

if and only if

$$\boxed{14p + 4q \leq 5.} \tag{51}$$

Proof. Branches 1, 3 preserve p , and branch 0 maps p to $7p/8$. Moreover $p_4 \leq p_2$, so only branch 2 can be critical.

If $4p + q \leq 1$, then $g_1 = 0$ and

$$p_2 = \frac{10p + 3q}{16} < \frac{1}{4}$$

under the same constraint. If $4p + q > 1$, then $g_1 = 4p + q - 1$, so

$$p_2 = \frac{14p + 4q - 1}{16}.$$

Thus $p_2 \leq 1/4$ is exactly (51). In the first region (51) holds automatically, so the equivalence is global. $\square$

If one asks that the children themselves satisfy (51), only the active branches 0, 2, 4 matter. Direct substitution gives the next three exact conditions:

$$57p + 14q \leq 20, \tag{52}$$

$$78p + 41q + 6r + 7g_1 + 2g_2 \leq 40, \tag{53}$$

$$70p + 42q + 4r + 14g_1 + 4g_2 \leq 40. \tag{54}$$

This exposes the recursive problem as an explicit prefix hierarchy rather than an implicit transport linear program.

8 Waring/Beta phase carriers

The physical top-tail choice is sharp for one step, but recursive closure benefits from a basis that remembers how mass arrived at the boundary. The following family provides such a phase coordinate.

Definition 8.1 (Waring-type phase family). For $k \geq 0$, let V_k be the positive-integer-valued law with

$$\mathbb{P}(V_k \geq j) = \frac{k+1}{k+j}, \quad j \geq 1, \tag{55}$$

and define

$$W_{r,k} := \text{Law}(r + V_k), \quad r \geq 0.$$

Lemma 8.2 (Waring aging identity). *For every $k \geq 0$,*

$$\boxed{\text{Law}(V_k - 1) = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}V_{k+1}.} \tag{56}$$

Consequently

$$\boxed{\text{DW}_{0,k} = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}W_{0,k+1},} \tag{57}$$

while for $r \geq 1$,

$$\text{DW}_{r,k} = W_{r-1,k}, \quad \text{UW}_{r,k} = W_{r+1,k}. \tag{58}$$

Proof. For $j \geq 1$,

$$\mathbb{P}(V_k - 1 \geq j) = \frac{k+1}{k+j+1} = \frac{k+1}{k+2} \frac{k+2}{k+j+1}.$$

The missing mass is $1/(k+2)$ at zero. The shifted identities are immediate. $\square$

There is also a continuous Beta-mixture representation. Let

$$g_t(j) = (1-t)t^{j-1}, \quad j \geq 1,$$

and $E_{r,t} := \delta_r * g_t$. Then

$$W_{r,k} = (k+1) \int_0^1 t^k E_{r,t} dt. \quad (59)$$

Indeed $\mathbb{P}(E_{r,t} \geq r+j) = t^{j-1}$, and integration gives (55). At the boundary,

$$DE_{0,t} = (1-t)\delta_0 + tE_{0,t}, \quad (60)$$

which makes the phase interpretation transparent: surviving a boundary hit multiplies the continuous component by t , while the leaked mass $1-t$ becomes boundary mass.

The critical root (23) is exactly

$$H_2 = W_{0,1}. \quad (61)$$

Definition 8.3 (Waring packet cone). Let $\mathfrak{W}$ be the cone of finite positive measures representable as

$$M = \sum_{j \geq 0} d_j \delta_j + \sum_{r,k \geq 0} w_{r,k} W_{r,k}, \quad d_j, w_{r,k} \geq 0, \quad (62)$$

with countable sums interpreted by monotone convergence whenever the total mass is finite. A *componentwise submeasure* is obtained by decreasing the coefficients $d_j, w_{r,k}$ without changing the component shapes.

Proposition 8.4 (Positive phase closure of the packet fan). *Suppose $B \in \mathfrak{W}$ is a probability law, $p = B(0) \leq 1/4$, and the packet G in Theorem 6.1 is chosen componentwise from a representation (62). Then every child B_x belongs to $\mathfrak{W}$.*

Proof. Equations (38)–(42) express each child as a positive linear combination of atoms and of A, G after applying only I, D, U . Equations (57)–(58) show that these operators preserve $\mathfrak{W}$. The explicit safe contribution from $p\delta_0$ produces only $\delta_0, \delta_1, \delta_2$, which already lie in the cone. $\square$

8.1 The exact first generation from the critical root

The first packet is empty because $H_2(0) = 0$. Hence the first generation is independent of any phase-selection policy and can be written exactly in the Waring basis.

Proposition 8.5 (First generation from H_2). *With $H_2 = W_{0,1}$, the active children are*

$$B_0 = \frac{7}{8}W_{0,1} + \frac{1}{8}W_{1,1}, \quad (63)$$

$$B_1 = B_3 = W_{0,1}, \quad (64)$$

$$B_2 = \frac{1}{16}\delta_0 + \frac{5}{8}W_{0,1} + \frac{1}{8}W_{0,2} + \frac{3}{16}W_{1,1}, \quad (65)$$

$$B_4 = \frac{1}{24}\delta_0 + \frac{7}{8}W_{0,1} + \frac{1}{12}W_{0,2}. \quad (66)$$

Consequently

child	$B_x(0)$	$B_x(1)$
0	0	7/24
1, 3	0	1/3
2	1/16	23/96
4	1/24	5/16

All three one-step Hall inequalities and the boundary face (71) hold strictly except for the trivial root value $p = 0$.

Proof. Set $p = 0$, $G = 0$, and $A = H_2$ in (38)–(42). The only nontrivial reduction is

$$DW_{0,1} = \frac{1}{3}\delta_0 + \frac{2}{3}W_{0,2},$$

which is (57) at $k = 1$. The displayed formulas follow immediately. The low-prefix values are obtained from $W_{0,k}(1) = 1/(k+2)$. $\square$

A useful structural distinction is already visible in (65)–(66). Branch 2 creates the interior phase component $W_{1,1}$ with mass $3/16 = 3B_2(0)$, whereas branch 4 creates no component with positive shift index r . This motivates treating the pure branch-4 ray separately.

8.2 Phase leak and greedy component selection

For a Waring component, the amount of mass sitting at reserve one is explicit:

$$W_{0,k}(1) = \frac{1}{k+2}, \quad W_{r,k}(1) = 0 \quad (r \geq 1). \quad (67)$$

Thus the phase index k has an exact operational meaning: after a boundary down-shift, a surviving component ages from k to $k+1$, and its next boundary leak decreases.

Given a representation (62), a componentwise packet of prescribed mass can be chosen by fractional knapsack: first use components with smaller reserve-one leakage, splitting the last coefficient if necessary. This minimizes $G(1)$ among packets that respect the chosen phase decomposition.

Lemma 8.6 (One-step leak bound). *Let $B \in \mathfrak{W}$ be a probability law, $p = B(0) \leq 1/4$, and $q = B(1)$. There exists a componentwise packet $G \leq B - p\delta_0$ of mass $3p$ such that*

$$\boxed{G(1) \leq \frac{3p}{1-p} q.} \quad (68)$$

Proof. The positive part has total mass $1-p$. The proportional componentwise packet

$$G_{\text{prop}} = \frac{3p}{1-p}(B - p\delta_0)$$

has mass $3p$ and satisfies equality in (68). A packet minimizing the reserve-one leakage over the same component representation can only improve this value. $\square$

Let

$$\eta := \frac{q}{1-p}.$$

Combining (68) with the exact packet formulas gives

$$B_2(0) \leq \frac{10p + 3\eta}{16}, \quad (69)$$

$$B_4(0) \leq \frac{5p}{8} + \frac{1+2p}{8}\eta. \quad (70)$$

Hence the boundary face

$$\boxed{0 \leq p \leq \frac{1}{4}, \quad 0 \leq \eta \leq \min\left(1, \frac{4-10p}{3}\right)} \quad (71)$$

is sufficient for one-step control of the zero mass in all active children. Indeed, (69) is at most $1/4$ under the second inequality in (71). For (70), the required bound is

$$\eta \leq \frac{2-5p}{1+2p}.$$

On $0 \leq p \leq 1/4$,

$$\frac{4 - 10p}{3} \leq \frac{2 - 5p}{1 + 2p},$$

because the difference after cross multiplication is

$$2 - 13p + 20p^2 = (4p - 1)(5p - 2) \geq 0.$$

For branch 0, the update is independent of the packet placement:

$$p_0 = \frac{7}{8}p, \quad \eta_0 = \frac{4p + 7\eta(1 - p)}{8 - 7p}. \quad (72)$$

Moreover branch 0 preserves the face (71). To see this, note that η_0 is increasing in η . On the sloping boundary $\eta = (4 - 10p)/3$, the desired inequality

$$\eta_0 \leq \frac{4 - 10p_0}{3}$$

is equivalent to

$$35p^2 + 48p - 16 \leq 0.$$

Its positive root is $4(-6 + \sqrt{71})/35 > 1/4$, so it holds throughout $[0, 1/4]$. On the cap $\eta = 1$,

$$\eta_0 = \frac{7 - 3p}{8 - 7p} \leq 1.$$

What remains is to use the Waring phase freedom to control the corresponding normalized low-prefix quantities on branches 2 and 4 simultaneously through all histories.

9 An all-depth theorem for the pure branch-4 ray

The branch-4 subsystem admits a closed proof. Let

$$W_k := W_{0,k}, \quad k \geq 1,$$

and consider probability laws of the form

$$B = p\delta_0 + \sum_{k \geq 1} w_k W_k, \quad w_k \geq 0, \quad \sum_{k \geq 1} w_k = 1 - p. \quad (73)$$

Write

$$q := B(1), \quad \eta := \frac{q}{1 - p}.$$

For a component W_k , set

$$a_k := W_k(1) = \frac{1}{k + 2} \in (0, 1/3],$$

and note that

$$W_k(2) = \frac{k + 1}{(k + 2)(k + 3)} = f(a_k), \quad f(a) := \frac{a(1 - a)}{1 + a}. \quad (74)$$

On $[0, 1/3]$, the function f is increasing and concave, since

$$f'(a) = \frac{1 - 2a - a^2}{(1 + a)^2} > 0, \quad f''(a) = -\frac{4}{(1 + a)^3} < 0.$$

Choose the packet $G = \sum g_k W_k$ of mass $3p$ by increasing leakage a_k , equivalently by decreasing k , splitting one coefficient if necessary. This is the greedy low-leak packet within the W_0 -cone.

Theorem 9.1 (Invariant branch-4 cone). *Suppose B has the form (73) and*

$$\boxed{0 \leq p \leq \frac{1}{7}, \quad 0 \leq \eta \leq \frac{1}{3}.} \quad (75)$$

Apply the packet fan with the greedy low-leak packet and condition on public output $x = 4$. Then the child B_4 again has the form (73) and again satisfies (75).

Proof. The class (73) is closed under branch 4: the only operators appearing in (42) are I and D , and

$$DW_k = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}W_{k+1}.$$

Thus no positive atom other than δ_0 is created.

Let

$$m := 3p, \quad g := G(1), \quad h := G(2), \quad r := B(2).$$

If $p > 0$, the greedy packet has average leakage no larger than the average leakage of the full positive part:

$$\frac{g}{m} \leq \eta. \quad (76)$$

For $p = 0$ the statement is trivial. By (74) and Jensen's inequality,

$$r \leq (1-p)f(\eta), \quad h \leq mf(g/m) \leq 3pf(\eta). \quad (77)$$

The exact prefix formulas on branch 4 give

$$p' = \frac{5p + q + g}{8}, \quad q' = \frac{7q + r + h}{8}. \quad (78)$$

Using $q = (1-p)\eta$, (76), and (77),

$$p' \leq \frac{5p + (1+2p)\eta}{8}, \quad (79)$$

$$q' \leq \frac{7(1-p)\eta + (1+2p)f(\eta)}{8}. \quad (80)$$

Since $\eta \leq 1/3$ and f is increasing,

$$f(\eta) \leq f(1/3) = \frac{1}{6}.$$

Therefore

$$p' \leq \frac{5p + (1+2p)/3}{8} = \frac{17p + 1}{24} \leq \frac{1}{7},$$

where the last inequality is exactly $p \leq 1/7$. Also

$$q' \leq \frac{7(1-p)/3 + (1+2p)/6}{8} = \frac{5-4p}{16}.$$

From the bound on p' ,

$$1 - p' \geq \frac{23 - 17p}{24}.$$

Hence

$$\eta' := \frac{q'}{1-p'} \leq \frac{3(5-4p)}{2(23-17p)} \leq \frac{1}{3},$$

because

$$9(5-4p) \leq 2(23-17p) \iff 0 \leq 1+2p.$$

This proves the invariant region. $\square$

Corollary 9.2 (The pure 4-ray is closed at every depth). *Starting from $H_2 = W_{0,1}$ and repeatedly taking public branch 4, the greedy Waring packet fan is well-defined for every depth and*

$$\boxed{B_{4^m}(0) \leq \frac{1}{7} < \frac{1}{4} \quad (m \geq 0).} \quad (81)$$

Moreover, from any state on this ray the first exit through branch 0 or 2 satisfies

$$B_{4^{m_0}}(0) \leq \frac{1}{8}, \quad B_{4^{m_2}}(0) \leq \frac{17}{112} < \frac{1}{4}. \quad (82)$$

Proof. At the root, $p = 0$ and $\eta = 1/3$, so Theorem 9.1 iterates indefinitely. For branch 0, $p_0 = 7p/8 \leq 1/8$. For branch 2, the exact formula and (76) give

$$p_2 = \frac{10p + 3q + g}{16} \leq \frac{10p + 3\eta}{16} \leq \frac{10/7 + 1}{16} = \frac{17}{112}.$$

□

This theorem removes the only active ray on which the packet has no interior $r \geq 1$ phase inventory. The unresolved part of the global conjecture is therefore a mixed-history regeneration problem after branches 0 and 2 have created shifted Waring components.

9.1 Finite-depth exact audit of the mixed recursion

For orientation, we record a finite exact enumeration for one deterministic componentwise low-leak policy. Packets are chosen by increasing value of $B(1)$ per unit component mass; ties among zero-leak components are broken by increasing shift index. All arithmetic is rational. Because branches 1 and 3 are identities, only words in $\{0, 2, 4\}$ create distinct states.

Depth	States	$p_{\max}$	$\eta_{\max}$	$p > 1/4$	face fail	Hall fail
0	1	0	0.333333	0	0	0
1	3	0.062500	0.326087	0	0	0
6	729	0.117624	0.315731	0	0	0
8	6561	0.115459	0.315365	0	0	0

At depth six the largest boundary mass occurs at the history 444442, where exactly

$$p = \frac{1748307947}{14863564800} = 0.1176237309\dots$$

The table is a finite certificate only; it is not used in the proof of Theorem 9.1 and does not replace the all-history argument still required below.

10 Resolvent boundary compression

The discounted Waring score of the preceding construction identifies the correct mechanism: a useful state statistic must remember both how far positive reserve lies from the hard boundary and how strongly it can leak when it returns. For a proof, however, a one-sided down-shift estimate is needlessly lossy at the boundary itself. The natural refinement is to sum all future discounted occupancies of the positive reserve. This produces a law-level resolvent and removes the phase-representation dependence from the projected recursion.

10.1 The resolvent leak scalar

Definition 10.1 (Positive-reserve resolvent). Fix $0 < \rho < 1$. For a finite positive measure M on $\{1, 2, \dots\}$, define

$$\mathcal{R}_\rho(M) := \sum_{j \geq 1} \rho^{j-1} M(j). \quad (83)$$

If $B = p\delta_0 + M$ is a probability law, write

$$q := M(1) = B(1), \quad R := \mathcal{R}_\rho(M). \quad (84)$$

The resolvent is a probability-generating-function statistic:

$$R = \frac{G_B(\rho) - p}{\rho}. \quad (85)$$

Thus, unlike the represented Waring leak score, R depends only on the reserve law itself.

Lemma 10.2 (Exact shift calculus). *Let M be supported on $\{1, 2, \dots\}$. Then*

$$\mathcal{R}_\rho(\mathbf{U}M) = \rho \mathcal{R}_\rho(M), \quad (86)$$

$$(\mathbf{D}M)(0) = M(1), \quad (87)$$

$$\mathcal{R}_\rho((\mathbf{D}M)|_{\{1, 2, \dots\}}) = \frac{\mathcal{R}_\rho(M) - M(1)}{\rho}. \quad (88)$$

Proof. All three identities follow by reindexing the defining series. In particular,

$$\sum_{j \geq 1} \rho^{j-1} (\mathbf{D}M)(j) = \sum_{j \geq 1} \rho^{j-1} M(j+1) = \frac{1}{\rho} \sum_{k \geq 2} \rho^{k-1} M(k).$$

□

In the geometric/Beta carrier basis of (59), the same scalar has the explicit form

$$\mathcal{R}_\rho(E_{r,t}) = \rho^r \frac{1-t}{1-\rho t}. \quad (89)$$

The factor $(1 - \rho t)^{-1}$ is the geometric resolvent of repeated future boundary exposure. This is the reason (88) is exact: the statistic counts not only the next possible leak, but the full discounted sequence of future positive-reserve occupancies.

10.2 Exact projected dynamics of the packet fan

Let a parent law be written as

$$B = p\delta_0 + M, \quad |M| = 1 - p,$$

and assume $p \leq 1/4$. Choose any packet

$$G \leq M, \quad |G| = 3p, \quad (90)$$

and put $A := M - G$. Besides the state variables (p, q, R) in (84), define the two packet observables

$$h := G(1), \quad g := \mathcal{R}_\rho(G). \quad (91)$$

For the second reserve layer write

$$r := M(2) = B(2), \quad s := G(2). \quad (92)$$

Theorem 10.3 (Exact resolvent packet recursion). *Under the packet fan of Theorem 6.1, branches 1 and 3 are identities. On the active branches the projected coordinates satisfy*

$$p_0 = \frac{7}{8}p, \quad q_0 = \frac{1}{2}p + \frac{7}{8}q, \quad (93)$$

$$p_2 = \frac{10p + 3q + h}{16}, \quad q_2 = \frac{4p + 10q + 3r + s}{16}, \quad (94)$$

$$p_4 = \frac{5p + q + h}{8}, \quad q_4 = \frac{7q + r + s}{8}. \quad (95)$$

Their positive-reserve resolvents are exactly

$$\boxed{R_0 = \frac{p}{2} + \frac{(7 + \rho)R - \rho g}{8}}, \quad (96)$$

$$\boxed{R_2 = \frac{p(4 + 2\rho)}{16} + \frac{1}{16} \left[\frac{3(R - q) + g - h}{\rho} + 10R + 3\rho R - \rho g \right]}, \quad (97)$$

and

$$\boxed{R_4 = \frac{1}{8} \left[\frac{R - q + g - h}{\rho} + 7R \right]}. \quad (98)$$

Proof. The boundary and reserve-one formulas follow by reading the coefficients of δ_0 and δ_1 in (38)–(42). For the resolvents, use (86)–(88) separately on A and G . Since

$$A(1) = q - h, \quad \mathcal{R}_\rho(A) = R - g,$$

branch 0 gives

$$8R_0 = 4p + 7(R - g) + \rho(R - g) + 7g = 4p + (7 + \rho)R - \rho g.$$

For branch 2,

$$\begin{aligned} 16R_2 &= p(4 + 2\rho) + 3 \frac{R - g - q + h}{\rho} + 10(R - g) + 3\rho(R - g) \\ &\quad + 4 \frac{g - h}{\rho} + 10g + 2\rho g, \end{aligned}$$

which simplifies to (97). Branch 4 is analogous:

$$8R_4 = \frac{R - g - q + h}{\rho} + 7(R - g) + 2 \frac{g - h}{\rho} + 7g,$$

and this is (98). □

The crucial point is that the infinite Waring inventory has disappeared from (96)–(98): for the boundary mass and the resolvent, the packet enters only through the two linear controls (h, g) .

10.3 Universal control and second-layer envelopes

The reserve-one and resolvent coordinates do not determine the entire law. The remaining dependence of q_2, q_4 in (94)–(95) is confined to the parent and packet masses at reserve two. Those masses admit explicit law-independent intervals.

Lemma 10.4 (Second-layer interval bounds). *Put*

$$T := 1 - p - q, \quad d := R - q. \quad (99)$$

Then

$$\max\left\{0, \frac{d - \rho^2 T}{\rho(1 - \rho)}\right\} \leq r \leq \min\left\{T, \frac{d}{\rho}\right\}. \quad (100)$$

Likewise, with

$$T_G := 3p - h, \quad d_G := g - h, \quad (101)$$

we have

$$\max\left\{0, \frac{d_G - \rho^2 T_G}{\rho(1 - \rho)}\right\} \leq s \leq \min\left\{T_G, \frac{d_G}{\rho}\right\}. \quad (102)$$

Proof. Since the mass of M above reserve one is T ,

$$d = \rho r + \sum_{j \geq 3} \rho^{j-1} M(j).$$

The remainder is nonnegative and at most $\rho^2(T - r)$. Hence

$$\rho r \leq d \leq \rho r + \rho^2(T - r),$$

which is exactly (100) after solving for r and intersecting with $0 \leq r \leq T$. The packet estimate is identical. $\square$

The upper endpoints in (100)–(102) immediately give the robust finite-dimensional bounds

$$q_2 \leq \frac{4p + 10q + 3r_+(p, q, R) + s_+(p, h, g)}{16}, \quad (103)$$

$$q_4 \leq \frac{7q + r_+(p, q, R) + s_+(p, h, g)}{8}, \quad (104)$$

where r_+ and s_+ denote the corresponding upper endpoints.

The packet-control pair itself satisfies the elementary necessary bounds

$$0 \leq h \leq \min\{q, 3p\}, \quad h \leq g \leq \min\{R, h + \rho(3p - h)\}. \quad (105)$$

More importantly, one has a universal existence estimate in every nonnegative linear direction.

Lemma 10.5 (Linear packet-selection bound). *For every $a, b \geq 0$, there exists a packet $G \leq M$ of mass $3p$ such that its controls (h, g) satisfy*

$$ah + bg \leq \frac{3p}{1 - p} (aq + bR). \quad (106)$$

Proof. Because $p \leq 1/4$, the factor $3p/(1 - p)$ lies in $[0, 1]$. The proportional packet

$$G_{\text{prop}} := \frac{3p}{1 - p} M$$

is therefore an admissible submeasure of mass $3p$ and has

$$G_{\text{prop}}(1) = \frac{3p}{1 - p} q, \quad \mathcal{R}_\rho(G_{\text{prop}}) = \frac{3p}{1 - p} R.$$

Thus it attains equality in the right-hand side of (106). A packet minimizing the linear cost $aG(1) + b\mathcal{R}_\rho(G)$ cannot be worse. Equivalently, the minimizing packet is a fractional knapsack with pointwise cost density

$$a \mathbf{1}_{\{j=1\}} + b \rho^{j-1}.$$

$\square$

Corollary 10.6 (Finite-dimensional robust control formulation). *Fix $0 < \rho < 1$. For the purpose of a sufficient all-history closure proof, every reachable node may be projected to the three-dimensional state*

$$\boxed{z = (p, q, R).}$$

A packet contributes only the two linear control observables

$$\boxed{u = (h, g).}$$

The active child coordinates (p_x, R_x) are given exactly by (93)–(95) and (96)–(98); the remaining child coordinate q_x is confined by the explicit interval envelope (100)–(104). In addition, the packet selector has the universal support-function estimate (106) in every nonnegative direction.

This is the principal compression obtained here. The recursion is no longer expressed as an evolution of countably many Waring coefficients. It is a robust finite-dimensional control problem: find a forward-invariant region in (p, q, R) , containing the projected critical root, and at every point choose a packet control (h, g) whose active children remain in that region. The identities above preserve the exact packet fan and therefore the original causal semantics.

The critical root is

$$H_2 = W_{0,1}, \quad p = 0, \quad q = \frac{1}{3}, \quad (107)$$

with

$$\boxed{R_*(\rho) := \mathcal{R}_\rho(H_2) = 2 \int_0^1 \frac{t(1-t)}{1-\rho t} dt.} \quad (108)$$

No approximation is involved in this initialization.

11 The remaining Waring packet invariance problem

We now isolate the single statement that would complete the logarithmic construction developed in this preprint.

Conjecture 11.1 (Packet invariance). *Start from*

$$B_\emptyset = H_2 = W_{0,1}.$$

At every reachable node h , write $p_h = B_h(0)$. There exists a packet submeasure

$$G_h \leq B_h - p_h \delta_0, \quad |G_h| = 3p_h,$$

such that the packet fan (38)–(42) can be iterated for every finite history and satisfies

$$\boxed{B_h(0) \leq \frac{1}{4} \quad \text{for every reachable history } h.} \quad (109)$$

Equivalently, the positive-reserve packet can always be selected so that the boundary compensation capacity $|G_h| = 3B_h(0)$ regenerates indefinitely.

The conjecture is deliberately formulated at the level actually used by the positive construction. It does not require a stationary child law, a fixed physical tail cutoff, or a history-independent scalar feedback. The Waring basis remains a useful structured carrier and yields the all-depth branch-4 theorem, but the resolvent formulation of Section 10 is law-level: the packet itself need only be a positive submeasure. Thus a proof may work directly with a finite-dimensional invariant-control region rather than preserve a particular phase decomposition at every history.

Theorem 11.2 (Conditional logarithmic upper bound). *If the packet invariance conjecture holds, then an infinite exact zero-repair tree exists with root H_2 , and*

$$\boxed{\mathcal{B}_n \leq 2H_{4n} = 2\log n + O(1).} \quad (110)$$

Combined with (9), this gives

$$\boxed{\mathcal{B}_n = \Theta(\log n).} \quad (111)$$

Proof. The conjecture recursively supplies a valid positive packet fan at every output history. By the causal lift, this defines an infinite exact tree with fresh iid Q -inputs and iid P -outputs. The root tail is (24); Theorem 3.1 then gives (110). The lower bound follows from (9). $\square$

12 Discussion

The constructive picture is now unusually explicit. The global logarithmic question has been reduced to a positive phase-selection problem inside a shift-stable cone:

1. the hard boundary creates a public-law defect of size $3p$;
2. a packet of positive reserve mass $3p$ cancels that defect exactly;
3. branches 1 and 3 are exact identity channels for the complete hidden law;
4. the top-tail packet exposes exact low-prefix recursions, including the sharp one-step criterion $14p + 4q \leq 5$;
5. the Waring basis turns a boundary down-shift into a positive aging operation, with leak $1/(k+2)$ and survivor phase $k+1$;
6. the critical root $H_2 = W_{0,1}$ already has the weak- L^1 tail required for the logarithmic bill.

The missing theorem is therefore not an asymptotic estimate and not an existence statement for a generic transport polytope. Section 10 replaces the countable phase inventory by a law-level robust control system. For a fixed $0 < \rho < 1$, the projected state is $z = (p, q, R)$, the packet enters through two linear controls $u = (h, g)$, the active p - and R -coordinates evolve exactly, and the only remaining second-layer uncertainty is confined by the explicit intervals (100)–(102). In every nonnegative control direction, Lemma 10.5 supplies a packet with a universal proportional cost bound. A forward-invariant safe region for this finite-dimensional robust system, containing the projected root H_2 , would close every history and complete the logarithmic upper bound. The preprint does not claim that invariant region as proved.

A Exact matrix audit for the packet fan

For completeness, we record the elementary identities behind Section 5. Multiplying by 64, the column sums of each of C^+ , C^0 , C^- are

$$(7, 20, 10, 20, 7) = 64Q.$$

The row sums are respectively

$$(11, 16, 16, 16, 5), \quad (8, 16, 16, 16, 8), \quad (7, 16, 16, 16, 9),$$

which are exactly $64R^+$, $64P$, $64R^-$. Thus

$$pR^+ + 3pR^- + (1 - 4p)P = P.$$

All nonzero cells of C^+ satisfy $x \leq y$, and all nonzero cells of C^0, C^- satisfy $x \leq y + 1$. Hence the construction is solvent on $J = 0$ and on $J \geq 1$, respectively.

B Notation cross-check

The Waring root $H_2 = W_{0,1}$ has

$$\mathbb{P}(H_2 \geq j) = \frac{2}{j+1}, \quad H_2(j) = \frac{2}{(j+1)(j+2)}, \quad j \geq 1,$$

and no atom at zero. Its truncated first moment is therefore

$$\mathbb{E} \min(H_2, N) = \sum_{j=1}^N \frac{2}{j+1} = 2(H_{N+1} - 1).$$

This is the exact quantity underlying the conditional estimate (110).

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